HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Led the **Computer System Validation (CSV)** process for a new **MES system** implementation, developing validation plans, URS, and IQ/OQ/PQ protocols.
- Executed **GxP system risk assessments** for automated manufacturing controls, identifying critical data integrity points and ensuring compliance with **EU GMP Annex 11**.
- Collaborated with **Engineering, IT, and Quality Assurance teams** to review system architecture and data flows for validation compliance, ensuring adherence to **21 CFR Part 11**.
- Developed and executed validation protocols for **PLC/SCADA systems** integrated with manufacturing equipment, documenting test results and managing deviations.
- Supported the validation lifecycle management for **enterprise business systems**, including change control, deviation management, and CAPA processes.
- Reviewed and approved validation documentation, including **Traceability Matrices and Validation Summary Reports**, ensuring alignment with GAMP 5 guidelines.
- Containerized a Python-based data processing service using **Docker** for deployment on an **AWS EC2** instance, improving deployment consistency.

SKILLS

Validation & Compliance – Computer System Validation (CSV), GAMP 5, GMP, Data Integrity, 21 CFR Part 11, EU Annex 11, ALCOA+, Risk Assessment, IQ/OQ/PQ, URS, Traceability Matrix, Validation Summary Report, Change Control, Deviation Management, CAPA

Systems & Technologies – MES, DCS, PLC/SCADA, ERP, SAP, OSIsoft PI, LIMS, BMS, DeltaV, Manufacturing Systems, Laboratory Systems, Automation Systems, Engineering Systems

Programming & Scripting – Python, Java, SQL, Shell Scripting, REST APIs

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases & Data Handling – PostgreSQL, MySQL, Oracle, Data Modelling, ETL, Power Query

Tools & Methodologies – Jira, Confluence, Agile/Scrum, Microsoft Office Suite (Word, Excel, PowerPoint)

PROJECTS

Validation of Manufacturing Execution System (MES)

- Developed comprehensive validation documentation including **Validation Plans, User Requirements Specifications (URS), and IQ/OQ/PQ protocols** for a new MES system.
- Performed **risk assessments** on system functionalities and interfaces, ensuring compliance with **GxP regulations** and data integrity principles.
- Coordinated with **Automation and Operations teams** to execute validation test cases and resolve identified deviations, ensuring successful system handover.

Automated Data Pipeline for Compliance Reporting

- Engineered a **Python ETL pipeline** to extract, transform, and load data from disparate sources into a central repository for compliance reporting.
- Implemented automated data validation checks using **SQL queries** against PostgreSQL, reducing manual data verification effort by approximately **30%**.
- Scheduled and monitored pipeline execution using **Linux cron jobs** and integrated with **Splunk** for enhanced logging and alerting on job failures.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)